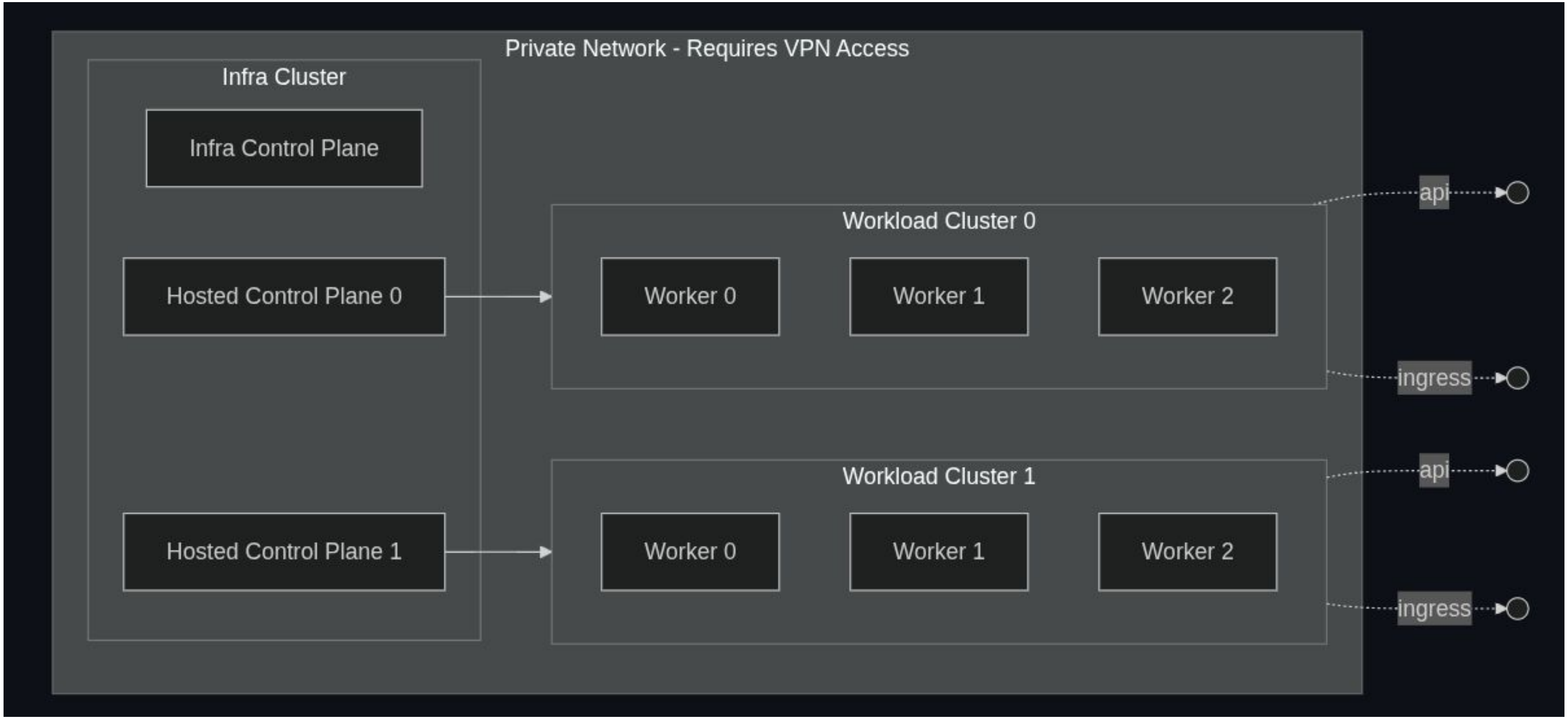
An Introduction to the Open Accelerator MOC Environment

Meera Malhotra
Open Accelerator Team
memalhot@redhat.com

Red Hat

Architecture Overview



A Look at Hardware

- Minimal cluster overview:
  - 20 GPUs, 1.28 TB GPU memory, and ~2 TB system memory/node
  - 3 FC830s, 3 H100s, and 2A100s

- Pure storage provides both file and object storage
  - Supports persistent storage for a range of workloads

CPU nodes

~ 2 TIB of memory and 6,047 GiB of RAM

H100 nodes

~ 1.5 TIB of memory and 80 GB HBM3 for VRAM

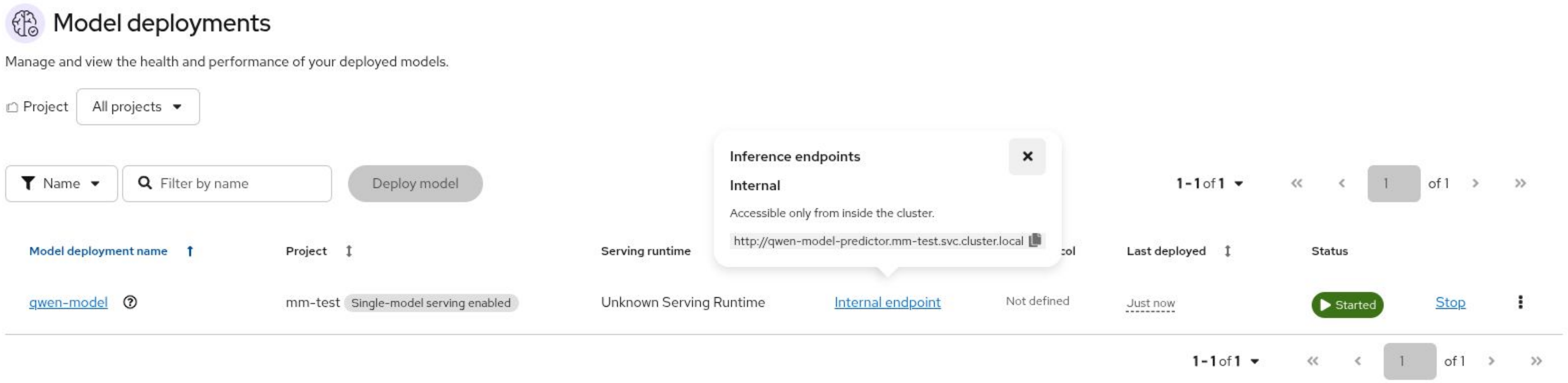
A100 nodes

~ 1 TIB of memory and 40 GB HBM for VRAM

What’s Possible

OpenShift allows for scale and industry grade AI deployment

- Model deployment and inference
- Pipelines for training
- Agent deployment with OpenShell guardrails
- VM offerings



Relevant Links

ArgoCD-managed configuration for the OAC OpenShift clusters:

https://github.com/CCI-MOC/oac-apps

Example workloads:

https://github.com/memalhot/open-science-test-cases

Test Harness:

https://github.com/asanaullah/ai-inference-uat

Demo:

https://github.com/asanaullah/oac-demo